

Let $f(x) = x^3 + x + 1$; the value of $f^{-1}(1)$ is

Select one:

☐ a. $\sqrt{3}$

☐ b. $-\sqrt{3}$

☒ c. 1

☐ d. 0

If $f(x) = \frac{3x+5}{2x+1}$ then

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a. $f^{-1}(x)$ is not a function ☐

$f^{-1}(x) = \frac{x+5}{2x-1}$.b ☐

~~$f^{-1}(x) = \frac{5-x}{2x-3}$.c ☐~~

$f^{-1}(x) = \frac{3x+5}{2x-1}$.d ☐

Suppose that M is a positive real number and that for all real numbers x a function f satisfies

$$-M \leq f(x) \leq M$$

Then $\lim_{x \rightarrow 0} x f(x) =$

Select one:

☐ a. ∞

☐ b. $-M$

☒ c. M

☐ d. 0

The value of the constant k that's make the function

$$f(x) = \begin{cases} \frac{x^2-16}{x-4}; & x \neq 4 \\ 2k; & x = 4 \end{cases}$$

continuous function.

Select one:

☐ a. $\frac{4}{3}$

☐ b. $\frac{8}{3}$

☒ c. 2

☐ d. 4

$$\lim_{x \rightarrow -2} \frac{\sin(x^2 - 3x - 10)}{x + 2}$$

Select one:

☐ a. -3

☐ b. 3

☒ c. -7

☐ d. 7

Let $f(x) = \frac{2x+1}{x^2+5x+2k}$.

The value of the constant k provided that $f(x)$ has one discontinuous point.

Select one:

☐ a. $\frac{9}{2}$

☒ b. $\frac{9}{2}$

☐ c. $\frac{25}{8}$

☐ d. $\frac{25}{8}$

$$\lim_{x \rightarrow \infty} \frac{3x^3 + 2x^2 - 12}{\pi + \pi x^3 + x} =$$

Select one:

☐ a. $\frac{2}{3}$

☐ b. Does not exist

☒ c. $\frac{3}{\pi}$

☐ d. 0

the vertical asymptote line for the function

$$f(x) = \begin{cases} \frac{x+2}{x^2-4}; & x > 1 \\ \frac{3x^2+1}{2x-1}; & x < 0 \end{cases}$$

Select one:

☒ a. $x = 2$

☐ b. $x = -2$ and $x = \frac{-1}{2}$

☐ c. $x = 2$ and $x = \frac{1}{2}$

☒ d. there's no vertical asymptote lines

If the limit

$$\lim_{x \rightarrow 3} \frac{x^2 + 2ax}{2x - 6}$$

exist, then the value of the constant a is

Select one:

☐ a. $\frac{3}{2}$

☒ b. -3

☒ c. 3

☐ d. $-\frac{3}{2}$

$$\lim_{x \rightarrow 0} \frac{4}{5x(\csc(3x))} =$$

Select one:

☒ a. $\frac{12}{5}$

☐ b. $\frac{3}{5}$

☐ c. 0

☐ d. $\frac{4}{5}$

If $f(x)$ is one to one function with domain $[1, 5]$ and range $[-9, 7]$ then the domain of $f^{-1}(3x - 2)$ is

Select one:

☐ a. $[-3, 5]$

☐ b. $[-6, 2]$

☒ c. $[-\frac{7}{3}, 3]$

☐ d. $[-\frac{11}{3}, \frac{5}{3}]$

$$\lim_{x \rightarrow 5} \frac{\sqrt{x-1}-2}{x-5} =$$

Select one:

☐ a. 4

☐ b. -4

☒ c. $\frac{1}{4}$

☐ d. $-\frac{1}{4}$

$$\lim_{x \rightarrow \infty} \csc\left(\frac{\frac{1}{4}x + 5}{2 - \frac{x}{4}}\right) =$$

Select one:

☐ a. $-\sqrt{2}$

☒ b. 2

☐ c. $\frac{-2}{\sqrt{3}}$

☐ d. $\sqrt{2}$

Let $f(x), g(x)$ be two function such that

$$f(x) = x^2 - 5 - 3x \text{ and}$$

$$(f + g)(x) = 2x^2 + 8x - 7$$

then $g(x) =$

Select one:

☒ a. $g(x) = x^2 + 5x - 2$

☐ b. $g(x) = x^2 + 11x - 2$

☐ c. $g(x) = x^2 + 5x - 12$

☐ d. $g(x) = x^2 + 11x - 12$

$$\lim_{x \rightarrow 6} \sqrt{\frac{x+2}{x-3}}$$

Select one:

☐ a. 0

☐ b. $\frac{8}{3}$

☒ c. $\sqrt{\frac{8}{3}}$

☐ d. Does not exist

The Horizontal Asymptote line for

$$f(x) = \sqrt{x^2 + 4x} - x$$

Select one:

- ☒ a. $y = 0$
- ☐ b. $f(x)$ has no Horizontal Asymptote
- ☐ c. $y = -2$ and $y = 2$
- ☐ d. $y = 2$

$$\lim_{x \rightarrow -2} \frac{x+3}{x^2-4}$$

Select one:

☐ a. 1

☒ b. ∞

☐ c. $-\infty$

☐ d. 0

One of the following graph is symmetric about Origin

Select one:

☐ a. $y = x^2 + x$

☒ b. $xy = 5$

☐ c. $x = 5y^2 + 9$

☐ d. $y = |x| + 5$

The inverse function for

$f(x) = x^2 - 2x$ where $x \geq 1$ is

Select one:

☒ a. $f^{-1}(x) = -\sqrt{x-1} + 1$

☐ b. $f^{-1}(x) = \sqrt{x+1} - 1$

☐ c. $f^{-1}(x) = -\sqrt{x+1} - 1$

☐ d. $f^{-1}(x) = \sqrt{x-1} + 1$

If the graph of $y = A\sin(Bx)$ has amplitude 3 and periodic 3 then

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$A = \pm \frac{2\pi}{3}$ a. $B = \pm 3$ and ☐

$B = -\frac{2\pi}{3}$ b. $A = -3$ and ☐

$B = \frac{2\pi}{3}$ c. $A = 3$ and ☐

$B = \pm \frac{2\pi}{3}$ d. $A = \pm 3$ and ☒

let c, k be two positive real number. If we translate the graph of $y = f(x)$ up k unit,
and translate it right c unit,
so we get the graph of

Select one:

☐ a. $y = f(x + c) - k$

☒ b. $y = f(x - c) + k$

☐ c. $y = f(x + c) + k$

☐ d. $y = f(x - c) - k$

the function $f(x) = x^3 - 4x + 1$ has a zero in the interval

Select one:

☐ a. $[-1, 0]$

☐ b. $[1, 2]$

☒ c. $[2, 3]$

☐ d. $[-4, -3]$

One of the following functions is odd function

Select one:

☐ a. $f(x) = \frac{1}{x^2} + x$

☒ b. $f(x) = \frac{1}{x^2} + x^5$

☐ c. $f(x) = \frac{1}{(x-1)^2} + \cos(x)$

☐ d. $f(x) = \frac{1}{x^2} + \cos(x)$